

Task: Workflow Design & Dataset Documentation for Virtual Try-On System

The objective of this task was to design the overall workflow for the Virtual Try-On (VTON) system and document publicly available fashion datasets useful for training, testing, and evaluating virtual try-on models. The work involved understanding different stages of the VTON pipeline, identifying suitable datasets, comparing their characteristics, and preparing documentation for project reference.

Virtual Try-On Workflow Design

The complete workflow of the virtual try-on system was designed to represent how an input image is processed and transformed into a final try-on output.

Workflow

User Image → Segmentation → Pose Detection → Garment Processing → Virtual Try-On → Final Output

Workflow Explanation

1. User Image

The process starts with an input image uploaded by the user. The image usually contains a person whose clothing appearance needs to be modified or tested with a selected garment.

2. Segmentation

Image segmentation is used to identify body regions and separate clothing from the background. This helps the model understand where garments should be placed and prevents unnecessary overlap during virtual try-on.

3. Pose Detection

Pose estimation detects body keypoints such as shoulders, elbows, waist, hips, and knees. These landmarks help align garments according to the posture and body orientation of the user.

4. Garment Processing

The selected garment image is processed before being applied to the user image. This stage may include resizing, masking, cloth segmentation, warping, and alignment based on the user's pose.

5. Virtual Try-On

At this stage, AI-based try-on methods combine user information and garment data to generate a realistic clothing fit. The garment is adapted to body position, orientation, and proportions.

6. Final Output

The final generated image shows the person wearing the selected garment with improved visual alignment and realistic appearance.

Dataset Documentation

Several publicly available datasets were studied for their usefulness in virtual try-on and fashion-related computer vision tasks.

A. DeepFashion

DeepFashion is a large-scale fashion dataset used for clothing recognition, attribute prediction, retrieval, and fashion understanding. Its features are:

- Large number of clothing images
- Clothing category labels
- Landmark annotations
- Consumer and shop image pairs

Useful for garment understanding, segmentation, clothing recognition, and fashion feature extraction.

Dataset Link: <https://mmlab.ie.cuhk.edu.hk/projects/DeepFashion.html>

B. DeepFashion2

DeepFashion2 improves upon DeepFashion by providing dense annotations and better clothing representation for segmentation and pose estimation. Its features are:

- 491K+ images
- 13 clothing categories
- Dense landmarks
- Segmentation masks
- Clothing pose and viewpoint variations
- Consumer-commercial clothing pairs

Helpful for clothing segmentation, pose estimation, cloth landmark extraction, and garment understanding for virtual try-on systems.

Dataset Link: <https://github.com/switchablenorms/DeepFashion2>

C. VITON Dataset

The VITON dataset is specifically designed for virtual try-on research and garment transfer experiments. Its features are:

- Person and clothing image pairs
- Clothing transfer setup
- Upper-body try-on focus
- Used in VTON model training

Useful for cloth alignment, garment transfer, and testing virtual try-on pipelines.

Dataset Link: <https://www.kaggle.com/datasets/xthanh2201/viton-dataset>

D. Dress Code Dataset

Dress Code is a high-resolution dataset developed for multi-category virtual try-on systems. Its features are

- High-resolution clothing images
- Multi-category garments
- Tops, dresses, lower-body clothing
- Better realism for try-on generation

Useful for advanced VTON experiments and testing garment fitting across multiple categories.

Dataset Link: <https://github.com/aimagelab/dress-code>

E. Fashionpedia

Fashionpedia is a fashion ontology dataset that combines clothing segmentation with fine-grained attribute recognition. Its features are:

- 27 apparel categories
- Fine-grained clothing attributes
- Segmentation masks
- Detailed fashion annotations

Useful for clothing attribute recognition, segmentation, and fashion understanding.

Dataset Link: <https://fashionpedia.github.io/home/index.html>

Progress Summary

During this task, the complete workflow of the virtual try-on system was documented and structured from input image processing to final output generation. Multiple fashion datasets were explored and compared to understand their role in virtual try-on applications. Documentation was prepared to organize dataset information, features, practical usage, and accessibility.

The study helped identify which datasets are more suitable for segmentation, pose estimation, garment processing, and virtual try-on model development. It also provided a clearer understanding of how data contributes to building realistic try-on systems.

Conclusion

This task focused on designing the overall workflow of a virtual try-on system and documenting publicly available datasets relevant to fashion AI and VTON applications. By studying datasets such as DeepFashion, DeepFashion2, VITON, Dress Code, and Fashionpedia, a better understanding was gained of dataset structure, annotations, and their importance in building realistic virtual try-on systems. The documentation prepared during this task will support future experimentation, implementation, and model development.